

Exploratory Data Analysis

This notebook presents an Exploratory Data Analysis (EDA) of the MVTec Anomaly Detection[1] dataset. MVTec AD is a benchmark dataset for unsupervised anomaly detection in industrial inspection and contains over 5,000 images across 15 object categories. Each category contains defect-free training images and a test set consisting of both normal and anomalous samples.

The purpose of this analysis is to:

1. Identify the object and texture categories present within the dataset.
2. Examine the distribution of normal and anomalous samples.
3. Investigate the defect types available within each category.
4. Quantify the spatial extent of anomalous regions using the provided segmentation masks
5. Analyse image characteristics such as resolution, brightness, sharpness, and contrast.
6. Identify potential challenges and limitations that may influence anomaly detection performance.
7. Inform preprocessing and model design decisions for the subsequent stages of the project.

1. Object and Texture Categories

All 15 available categories that are present within the MVTec dataset.

bottle	cable	capsule	carpet	grid
hazelnut	leather	metal_nut	pill	screw
tile	toothbrush	transistor	wood	zipper

2. Normal and Anomalous Sample Distribution

This section examines how samples are distributed across the dataset. Understanding the balance between normal and anomalous images helps identify class imbalances that may influence anomaly detection performance and evaluation metrics.

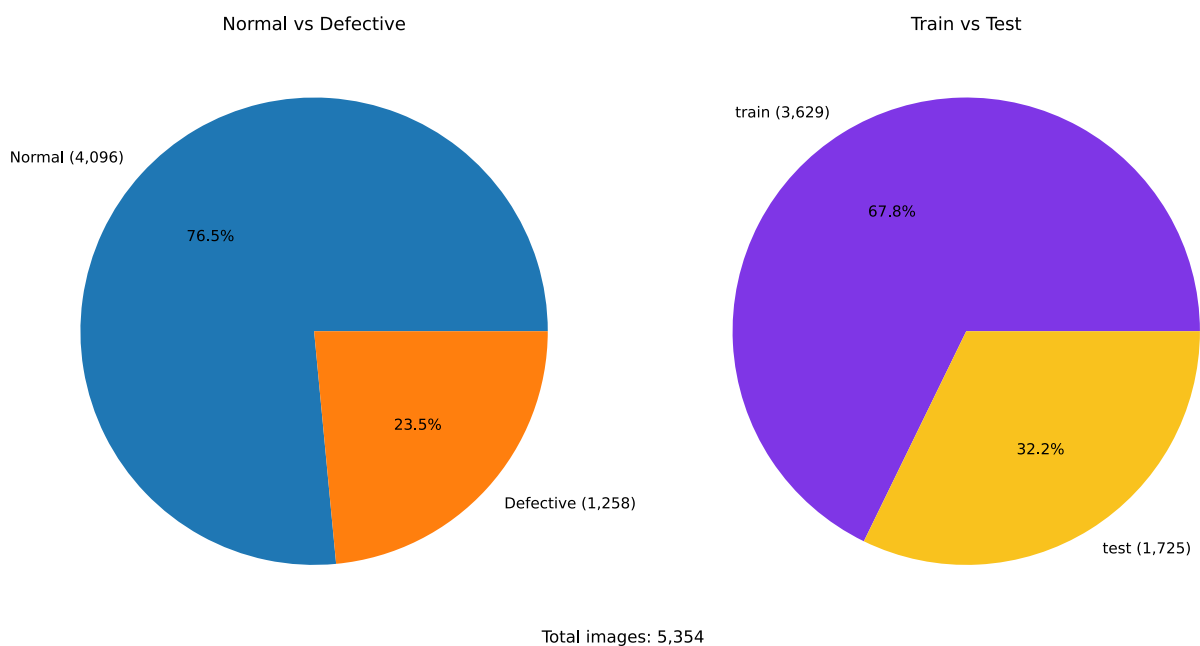


Figure 1: Overall normal/defective and train/test distribution of MVTec AD

2.1. Category Distribution

This investigates how images are distributed across the 15 MVTec categories. Identifying category-level imbalances helps determine whether certain categories may be overrepresented during evaluation.

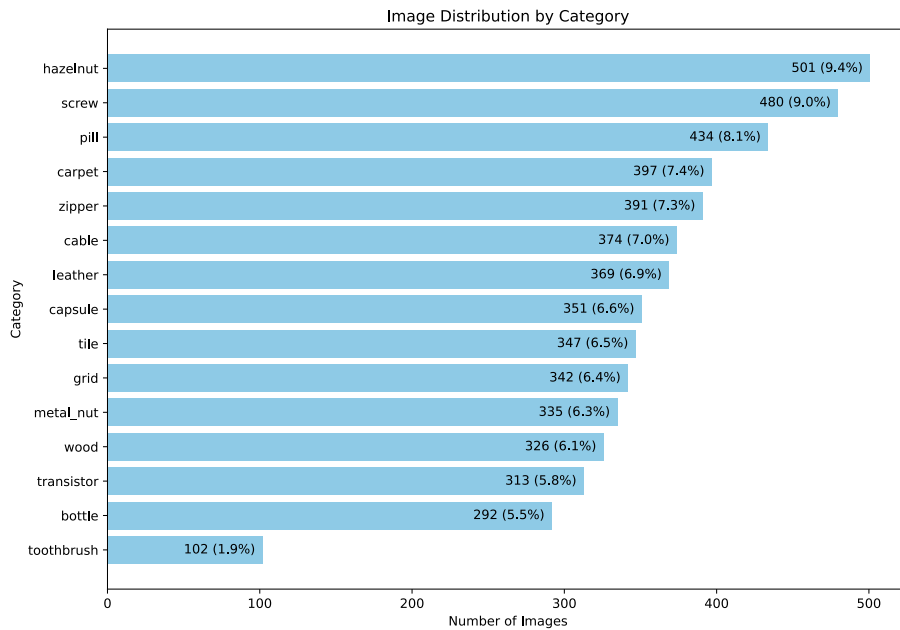


Figure 2: Distribution of images across MVTec AD categories

The dataset is reasonably balanced across categories, although some categories contain more images than others. No category dominates the dataset to a degree that would significantly bias overall evaluation.

This figure shows the proportion of normal and defective samples within each category. As MVTec is designed for anomaly detection, categories may contain different ratios of normal and anomalous images depending on the number of defect types available.

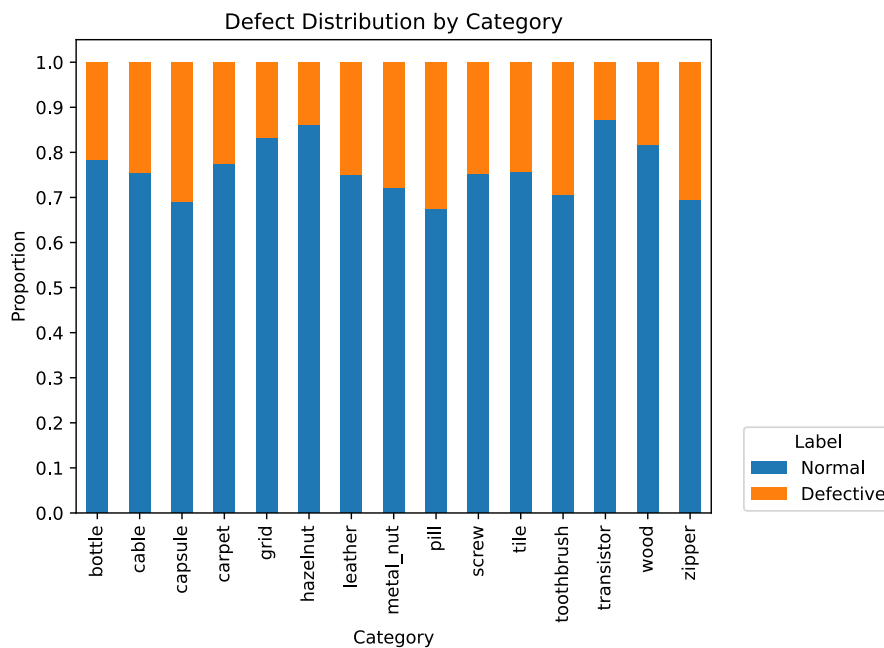


Figure 3: Proportion of normal and defective samples by category

The proportion of defective samples varies between categories. Categories containing a larger number of defect types generally show a higher proportion of anomalous test samples.

3. Defect Types by Category

Understanding the variety of defect types present within each category is important for assessing dataset complexity. Categories containing a larger number of defect types may present a more challenging anomaly detection task.

The following visualisations show how defect types are distributed within each category's test set. This provides insight into the relative frequency of different anomaly types.

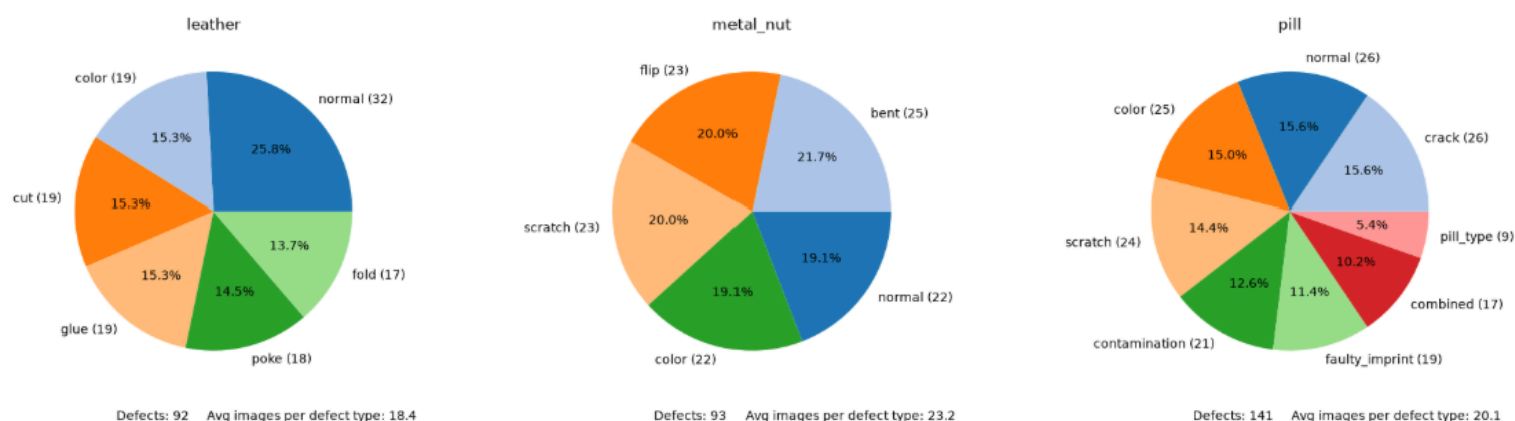


Figure 4: A sample of the distributions of defects in the data

The number and distribution of defect types varies substantially between categories. Some categories contain only a small number of anomaly classes, while others contain a wider range of defect variations.

Examples of normal and defective samples are shown below to illustrate the visual variation between anomaly types. A larger number of certain defect types may indicate increased anomaly diversity and greater detection complexity.

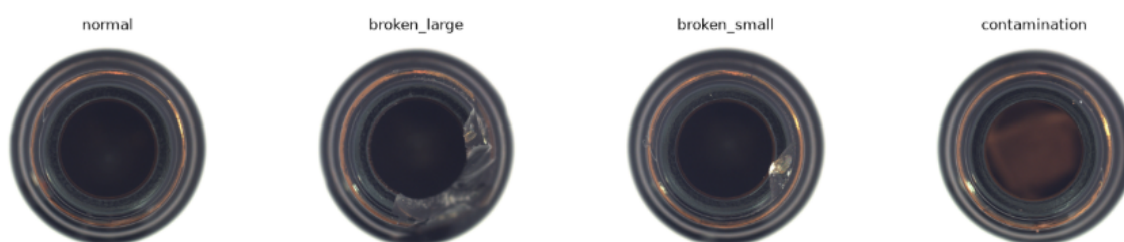


Figure 5: Example of defects in the bottle category

3.1. Defect Size and Image Coverage

Defect severity is not determined solely by defect type, the spatial coverage of an anomaly may also influence detection difficulty. Using the provided segmentation masks, the proportion of each defective image occupied by anomalous regions was calculated to examine how defect size varies both within and between categories.

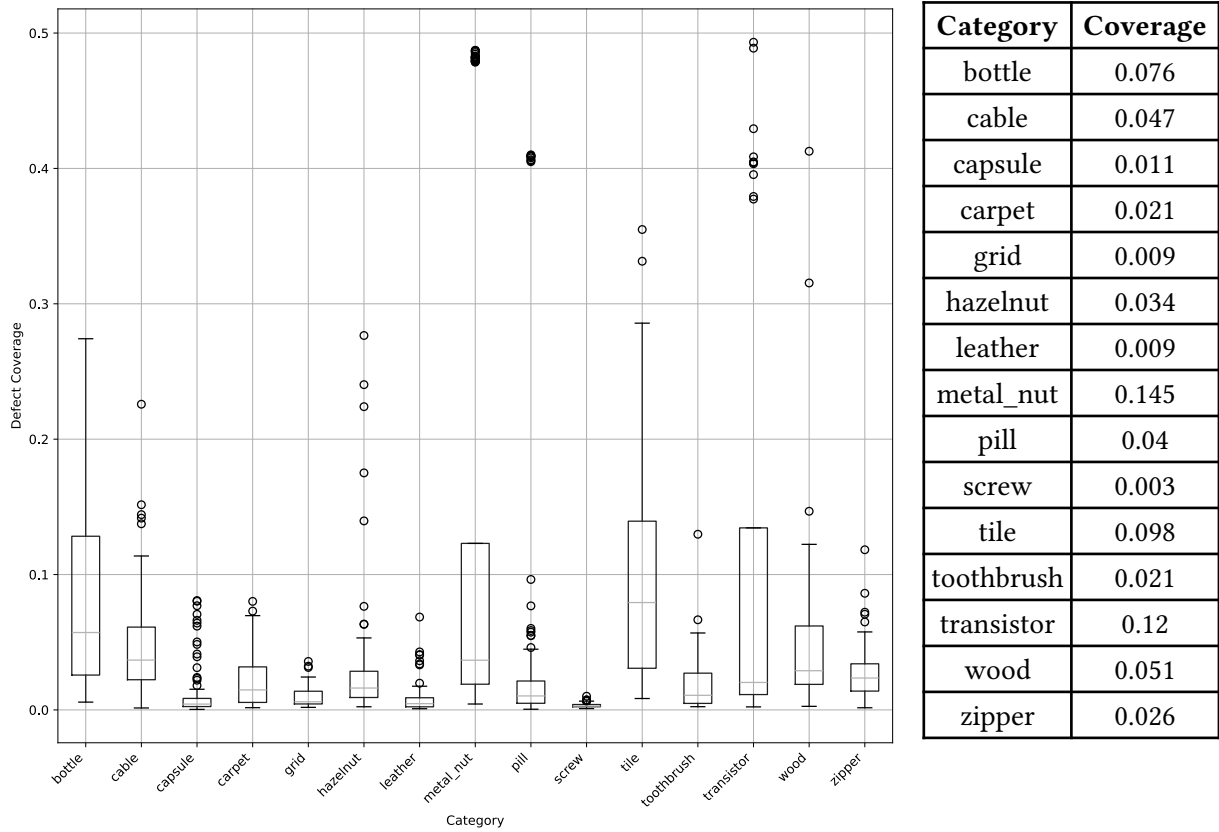


Figure 6: Defect coverage by category accompanied by the average coverage by category.

As shown in Figure 6, defect coverage varies substantially both between and within categories. Metal nut is particularly notable: while many defective samples occupy approximately 10% of the image, several contain anomalous regions approaching 50%. As shown in Figure 4, one of the metal nut anomaly types consists of a flipped object, which may contribute to the substantially larger spatial coverage observed for some samples. Further inspection shows that these high-coverage samples correspond to the flip defect type, in which a substantial portion of the object is spatially altered. This explains the unusually large anomalous regions observed for the category and highlights how defect type can strongly influence spatial coverage. In contrast, screw contains particularly small anomalous regions, with defective samples covering only 0.3% of the image on average. Such limited spatial coverage may make this category more difficult to detect, particularly when using global image representations, where the anomalous region contributes only a small proportion of the overall representation.

4. Image Characteristics

This section investigates several image characteristics that may influence feature extraction and anomaly detection performance. Resolution, brightness, sharpness, and contrast are analysed to identify potential inconsistencies or preprocessing requirements.

Image-level statistics are computed for every image in the dataset. Brightness is measured as the mean pixel intensity, contrast as the standard deviation of pixel intensities, and sharpness using the variance of the Laplacian.

Category	Brightness		Sharpness		Contrast	
	mean	std	mean	std	mean	std
bottle	136.74	1.71	44.38	4.63	92.72	0.95
cable	102.06	4.71	46.01	10.6	50.77	2.81
capsule	172.24	4.32	17.25	3.37	63.0	1.68
carpet	91.06	2.48	300.72	56.27	35.39	1.54
grid	114.11	9.23	30.09	11.4	39.18	7.52
hazelnut	51.72	4.13	43.01	3.84	26.49	5.13
leather	80.39	6.91	46.96	18.76	11.75	2.09
metal_nut	58.49	4.39	62.75	14.18	43.25	4.58
pill	76.33	4.96	142.1	96.86	78.45	4.05
screw	183.61	6.35	23.41	2.13	33.42	1.98
tile	116.35	8.45	46.61	14.97	31.5	2.23
toothbrush	47.02	8.02	167.17	137.77	55.81	1.61
transistor	79.52	3.44	63.58	8.43	42.07	1.56
wood	135.9	7.2	89.45	24.39	16.09	3.04
zipper	101.97	2.02	72.46	9.66	82.33	1.23

Table 1: Image-level statistics by MVTec AD category.

5. Observations

Several notable observations can be made from the image characteristic analysis:

- Image resolution is consistent within each category but varies between categories, ranging from 700x700 to 1024x1024 pixels.
- Brightness varies significantly between categories due to differences in object appearance and material properties, but remains highly consistent within individual categories.
- Sharpness and contrast exhibit greater variation between categories, reflecting differences in texture complexity and surface structure.
- Categories such as carpet, wood, and toothbrush contain substantially more texture detail than smoother categories such as capsule and screw.

These findings suggest that resizing and normalization will be necessary during preprocessing, while category-specific texture complexity may influence anomaly detection performance.

Bibliography

- [1] M. Software, “The MVTec Anomaly Detection Dataset,” [Online]. Available: <https://www.mvtec.com/research-teaching/datasets/mvtec-ad>

Baseline Model

This section explores the pretrained embeddings produced by the DINOv2 ViT-S/14 model. DINOv2 was selected because its self-supervised training produces general-purpose visual representations without requiring task-specific labels, making it well suited to the unsupervised setting considered

in this work. The ViT-S/14 (approximately 21 million parameters) variant was chosen as a relatively compact model that provides both global CLS and local patch representations, allowing the same backbone to support the image-level experiments in this study and subsequent investigation of local anomaly representations.

The purpose of this section is to:

1. Evaluate several anomaly scoring methods using the extracted CLS embeddings:
 - Centroid Distance
 - K-Nearest Neighbours (KNN)
 - Mahalanobis Distance
2. Analyse the distribution of anomaly scores for normal and defective samples to determine how well each method separates the two classes.
3. Compare methods using AUROC to identify the most effective baseline anomaly scoring approach. AUROC is used because it evaluates ranking performance across all possible decision thresholds, avoiding the need to select a fixed anomaly threshold for baseline comparison.
4. Identify categories that are already well separated within the embedding space, as well as categories that remain challenging and may benefit from additional experimentation.

The results obtained in this section serve as a baseline against which subsequent approaches, including fine-tuned and semi-supervised representations, are compared.

6. Evaluate several anomaly scoring methods using the extracted CLS embeddings

Metrics chosen include:

- **Centroid Distance:** Computes the Euclidean distance between each test embedding and the centroid (mean embedding) of the corresponding training category.
- **KNN:** The training embeddings are used as the reference set for KNN. Each test embedding is then compared against them to find its K nearest training embeddings and their distances. The anomaly score is taken as the distance to the closest neighbour.
- **Average KNN:** Computes the average distance between each test embedding and its K (K=5) nearest training embeddings.
- **Mahalanobis Distance:** Computes the distance between each test embedding and the training distribution while accounting for the covariance structure of the embedding space.

7. Analyse the distribution of anomaly scores for normal and defective samples

A reusable plotting function is used to compare the anomaly score distributions of normal and defective test samples for each category. The score distributions are plotted separately for each anomaly scoring method, allowing the degree of separation between normal and defective samples to be visually inspected.

AUROC is also computed for each category to quantify how well the selected scoring method ranks defective samples above normal samples.

Method	mean	median	std	Min Category	Value	Max Category	Value
Centroid	0.845	0.849	0.135	screw	0.506	leather	0.997

KNN	0.923	0.933	0.06	screw	0.804	leather	1.0
Avg-KNN	0.92	0.926	0.062	screw	0.786	leather	1.0
Mahalanobis	0.958	0.972	0.049	screw	0.802	tile	1.0

AUROCS for each scoring method

The centroid-based method has reasonable performance across several categories; however, there is a noticeable overlap between the score distributions of normal and defective samples. This indicates that the distance from the centroid alone is not enough to fully describe the embedding space. While the centroid gets the average location of normal samples, it does not take into account the spread or covariance of the distribution, meaning some anomalous samples receive scores similar to normal samples.

The K-Nearest Neighbour approach improves upon the centroid-based method by producing clearer separation between the score distributions of normal and defective samples. The leather category has perfect separation, achieving an AUROC of 1.0. This suggests that anomalous samples are characterised more by their distance to nearby normal samples than by their distance to a single global centroid. The results indicate that normal and defective samples occupy distinct regions of the embedding space, allowing KNN to identify anomalies through local neighbourhood structure rather than relying solely on the overall centre of the distribution.

Average KNN does not improve upon the performance of standard KNN and achieves a lower average AUROC across the dataset. This suggests that the nearest normal neighbour contains the most informative signal for anomaly detection. By averaging across multiple neighbours, the anomaly score becomes influenced by additional normal samples that may lie further away in the embedding space, reducing the contrast between normal and defective samples. As a result, the score distributions exhibit greater overlap, leading to weaker separation and reduced anomaly detection performance.

The Mahalanobis score performed best among all evaluated methods, achieving perfect separation between normal and defective samples in several categories, including bottle, leather, and tile. Mahalanobis achieved the highest mean AUROC overall. KNN performed marginally better on screw, achieving 0.804 compared with 0.802 for Mahalanobis. Overall, Mahalanobis was therefore selected as the preferred anomaly scoring method for this study.. Furthermore, the categories exhibiting perfect separation could be used as control groups in future experiments to evaluate whether proposed modifications improve anomaly detection performance or instead degrade the structure and separability of the embedding space.

Despite strong overall performance, categories such as screw achieved lower AUROC values across multiple scoring methods. This suggests that normal and defective samples occupy more similar regions of the embedding space, making anomaly detection more challenging. These categories provide useful benchmarks for evaluating future improvements, as successful methods should increase separability in these difficult cases.

Embedding Space Visualisation

This section investigates how embeddings extracted from the model are organised within the embedding space.

The embedding space is analysed using three dimensionality reduction techniques, including:

- PCA
- t-SNE
- UMAP

Analysis is performed at both the image level, using CLS embeddings, and the local level, using patch embeddings. This allows investigation of how entire images relate to one another within the embedding space, as well as how individual patches relate to other patches within the same image.

Scoring metrics will be used to analyse cluster quality and separation, including:

- Silhouette Score (Sil.)
- Davies-Bouldin Index (DB)
- Calinski-Harabaz Index (CH)
- Intra-Cluster Distance
- Inter-Cluster Distance

(Should probably explain metrics just so its giving a why these ones are chosen, same with what does each tell and why, this is a dissertation so it can go more repetitive)

The objective of this notebook is to understand the structure of the embedding space and identify characteristics that can help with anomaly detection. These findings will be used to explain the performance of the baseline model and provide a reference for future experiments involving fine-tuning and alternative embedding representations.

CLS Embedding Projections

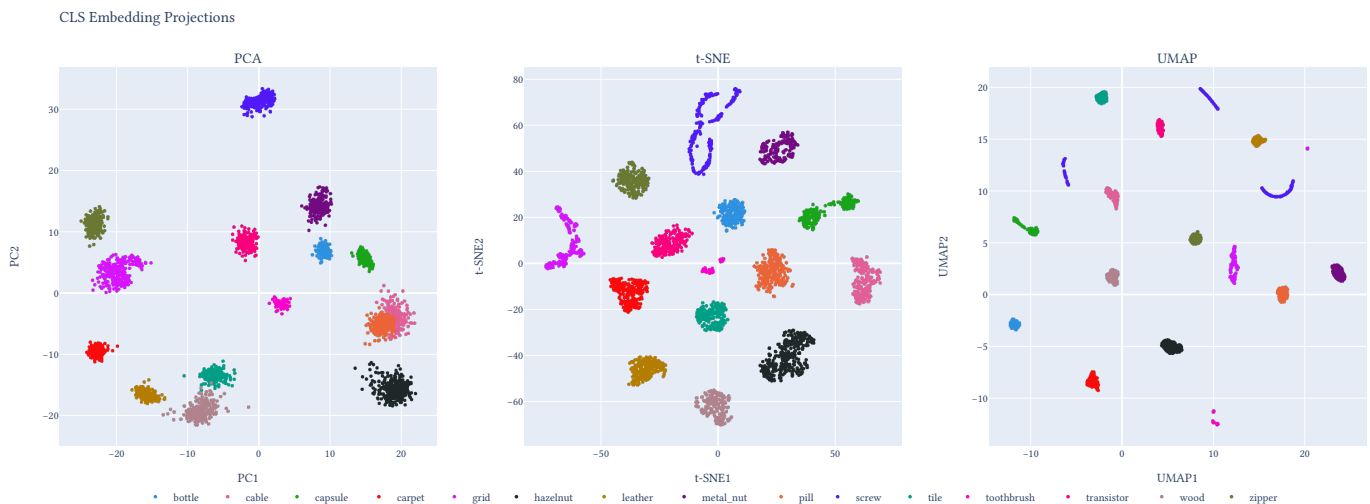


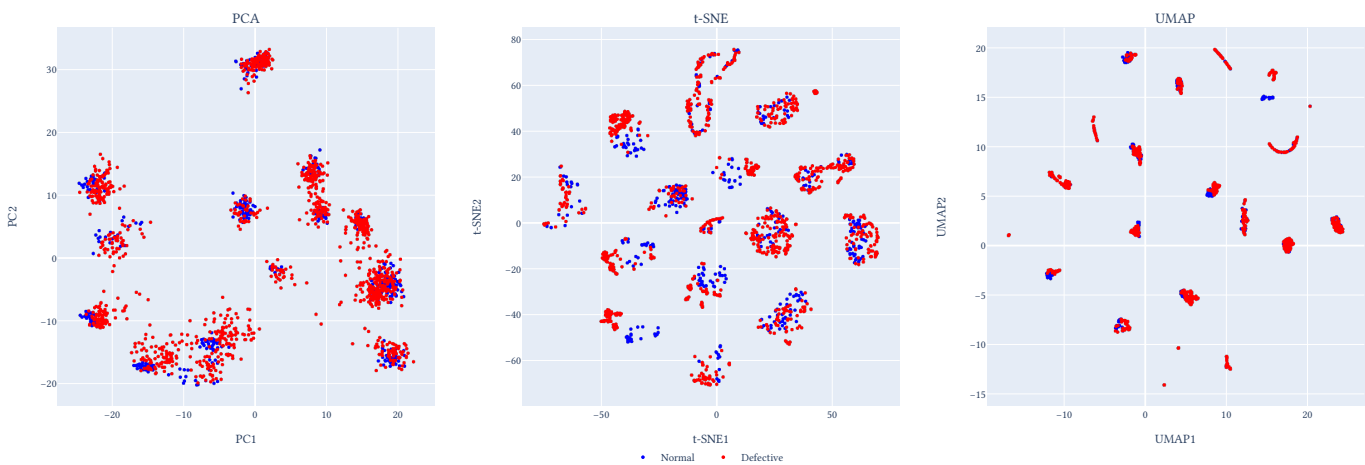
Figure 7: Embedding Projections of CLS tokens

Within the PCA projection, we can observe how the embedding space is organised along the two principal components that capture the greatest variance in the data. Several categories form distinct clusters that reflect similarities in their visual appearance. For example, relatively flat and homogeneous objects such as tile, leather, and wood are positioned closer together, while categories such as screw occupy a more distant region of the embedding space due to their more complex geometric structure and visual texture.

In the t-SNE projection, we can observe how compact or diffuse each category is within the embedding space. Most categories form relatively tight clusters, indicating that samples from the same category are represented consistently by the model. Notable exceptions include grid and screw, which exhibit more diffuse and fragmented structures. The screw category is particularly interesting, as it also achieved some of the lowest anomaly detection performance in the previous analysis. Rather than forming a single compact cluster, the embeddings are distributed in a figure-eight like structure, suggesting that normal and defective samples may occupy multiple overlapping regions of the embedding space. This reduced compactness may contribute to the lower AUROC scores observed for this category, as anomalous samples become more difficult to distinguish from normal samples using anomaly scoring methods.

Finally, the UMAP projection highlights how distinct the local neighbourhoods are within the embedding space and how compact these neighbourhoods remain. The observations regarding cluster compactness are similar to those seen in the t-SNE projection, with grid and screw again emerging as the most diffuse categories. Rather than forming a single coherent cluster, samples from these categories occupy multiple regions of the embedding space, with the screw category appearing to split into three distinct subregions. This provides further insight into the lower anomaly detection performance observed for screw in the previous analysis. Since normal samples are distributed across several disconnected regions rather than a single compact cluster, anomaly scoring methods may struggle to distinguish normal and defective samples consistently, resulting in reduced AUROC performance.

CLS Embedding Projections(Normal vs Defective)



For the PCA projection, there is unsurprisingly little separation between normal and defective samples. Since PCA preserves the directions of greatest global variance, the principal components are dominated by category-level differences rather than the subtle differences introduced by anomalies. As a result, normal and defective samples largely occupy the same regions of the embedding space.

Within the t-SNE projection, we can observe how defective samples are positioned relative to local neighbourhoods of normal samples. For many categories, particularly those containing flat and visually consistent objects, normal and defective samples form relatively distinct regions. However, the screw category remains an exception. Here, normal and defective samples occupy almost identical regions of the embedding space, with little evidence of a distinct anomaly cluster. This provides a possible explanation for the lower anomaly detection performance observed previously, as the embedding representation does not naturally separate normal and defective screw samples.

The UMAP projection exhibits similar behaviour to t-SNE. For many categories, defective samples tend to occupy regions adjacent to, but partially separated from, normal samples. However, in more challenging categories such as screw, normal and defective samples continue to share the same embedding regions. This suggests that the model struggles to learn discriminative representations for these anomalies, making them difficult to separate using distance-based anomaly scoring methods.

CLS Embedding Statistics

Metric	Mean	Median	Std	Min	Max	Worst	Best
Cluster Quality							
Sil.	0.0401	0.0425	0.0588	-0.0722	0.1615	capsule	leather
DB	3.2804	2.8931	1.4754	1.5058	6.6545	screw	leather
CH	8.8111	6.6562	6.7326	2.6555	27.5506	screw	leather
Inter-Cluster Distance							
Normal	13.5318	12.1777	4.0829	7.3152	21.3277	wood	bottle
Defect	20.5365	19.4394	5.0035	13.0549	30.4626	capsule	tile
Sep. Ratio	1.5956	1.3713	0.45	0.9871	2.4991	screw	bottle
Intra-Cluster Distance							
Normal	12.6352	11.4615	3.9479	6.9367	18.9459	wood	bottle
Defect	17.7923	16.6142	3.7904	12.5704	25.5036	capsule	tile

The mean Silhouette Score across categories is 0.04, indicating weak separation between normal and defective CLS embeddings. Since a score close to zero indicates substantial overlap between clusters, the result suggests that normal and defective samples often occupy similar regions of the CLS embedding space. This is consistent with the earlier EDA(get table reference), which showed that several categories contain highly localised anomalous regions. Because the CLS token summarises information across the entire image, these small local changes may contribute only weakly to the global representation, causing normal and defective embeddings to remain similar.**(make sure not to repeat this too many times)**

Defect Inter-Cluster Distance is consistently larger than Normal Inter-Cluster Distance, suggesting that defective samples tend to lie further from the category centroid than normal samples. This raises the question of whether the magnitude of this separation is associated with anomaly detection performance, which can be investigated by comparing category-level separation ratios with AUROC.

The mean Separation Ratio of 1.60 indicates that defective samples lie approximately 60% farther from the category centroid than normal samples, providing a measurable distance-based distinction between the two groups. This suggests that the pretrained CLS embeddings already contain a useful anomaly signal despite no task-specific training.

Normal Intra-Cluster Distance is generally lower than Defect Intra-Cluster Distance, indicating that normal samples form more compact clusters while defective samples exhibit greater variability. This may partly reflect the diversity of defect types within each category, since different anomalies can alter the visual representation in different ways. The earlier EDA also showed substantial variation in defect coverage within some categories, suggesting that defective samples can differ considerably in both appearance and spatial extent. This additional variation may contribute to the greater spread of defective embeddings around their centroid.

The category-level statistics also highlight screw as a consistently challenging category. It achieves the worst Davies-Bouldin and Calinski-Harabasz scores among the CLS embeddings, indicating weaker cluster separation and less coherent structure. This agrees with the earlier t-SNE and UMAP projections^{Figure 7}, where screw formed diffuse and fragmented regions rather than a single compact cluster. The same pattern is reflected in the baseline anomaly-detection results, where screw produced the lowest AUROC for several scoring methods, further suggesting that its normal and defective representations are difficult to separate.

Patch Embedding Statistics

The silhouette scores are generally much higher than those seen for the CLS embeddings, with most categories falling between 0.1 and 0.3. This suggests that patch embeddings form more coherent local structures than global image-level embeddings. (Discuss more than just saying its better and also reference that previous table)

Cluster Quality Metrics			
Category	Sil.	DB	CH
bottle	0.1032	2.0161	21.0375
cable	0.1234	2.0764	17.4583
capsule	0.1103	1.4781	8.0617
carpet	0.2666	1.1308	25.0733
grid	0.2727	1.1495	22.83
hazelnut	0.1479	1.4992	19.9355
leather	0.2748	0.946	14.5154
metal_nut	0.0924	2.0189	31.2216
pill	0.0977	1.7748	14.5052
screw	0.1252	1.2913	5.5748
tile	0.2694	1.4006	56.9894
toothbrush	0.106	1.7726	13.9147
transistor	0.086	2.7955	13.6462
wood	0.2768	1.4002	40.1426
zipper	0.083	1.7521	12.1111

Inter-Cluster Distances			
Category	Normal	Defect	Sep. Ratio
bottle	35.567	38.7165	1.0888
cable	36.312	42.2481	1.1644
capsule	37.6489	45.1315	1.1989
carpet	28.0213	44.2434	1.5799
grid	29.8463	44.9653	1.5073
hazelnut	39.1841	47.6354	1.2157
leather	30.8956	49.6409	1.6115
metal_nut	37.5273	37.4852	0.9997
pill	37.9695	42.7971	1.1274
screw	37.5862	45.9468	1.2225
tile	32.41	45.3849	1.3964
toothbrush	37.6068	43.5274	1.1573
transistor	32.5668	37.1644	1.1422
wood	30.474	42.6916	1.4054
zipper	33.2571	37.3918	1.1236

Most categories achieve separation ratios above 1, suggesting that defects are generally positioned further from the normal embedding distribution. Leather (1.61) demonstrates the strongest separation, while metal_nut (≈ 1.00) exhibits almost no difference between normal and defective patch distances. This indicates that defective metal nut patches occupy nearly the same embedding regions as normal patches.

Most categories have an intra-cluster distance value close to 1, suggesting that defective patches are similarly dispersed to normal patches. However, categories such as leather (1.08) show defective patches becoming more diffuse, while wood (0.82) exhibits more compact defective clusters. This suggests that anomalies affect categories differently, with some defects introducing greater variation and others producing consistent visual patterns.

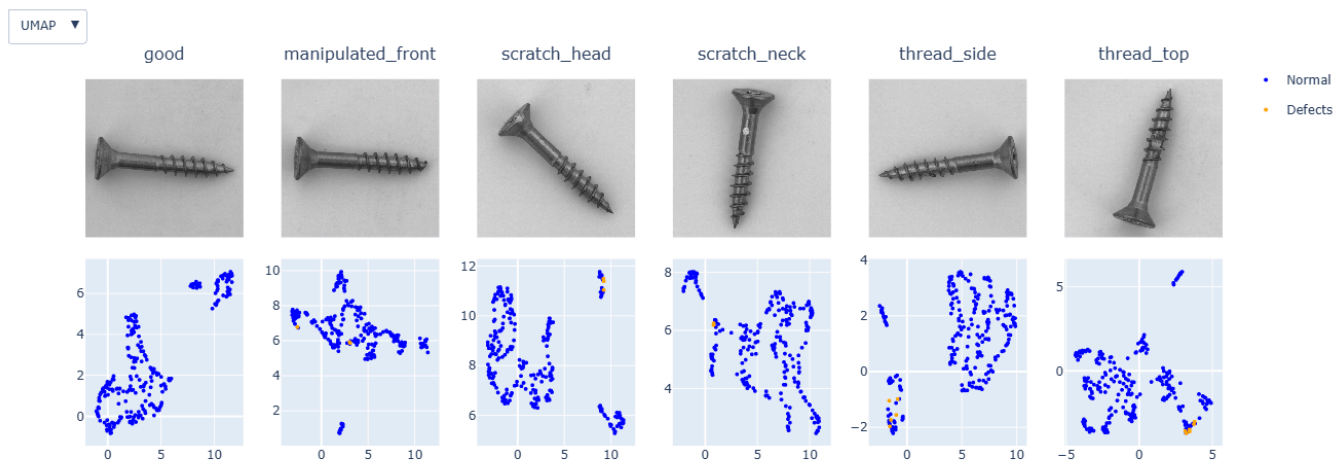
Intra-Cluster Distances		
Category	Normal	Defect
bottle	33.8505	24.6126
cable	34.2263	27.7112
capsule	37.0514	20.1856
carpet	24.8757	15.8434
grid	24.6649	16.2601
hazelnut	37.6479	23.3492
leather	26.4941	12.5874
metal_nut	34.3597	25.8242
pill	36.554	23.9242
screw	35.9715	17.2762
tile	25.9535	22.2262
toothbrush	36.3193	25.1748
transistor	30.3868	27.0359
wood	23.5885	20.3072
zipper	31.261	19.5223

Normal Intra-Cluster Distances

Category	Test	Train	Ratio
bottle	33.8505	34.6712	0.9763
cable	34.2263	35.7712	0.9568
capsule	37.0514	37.4239	0.99
carpet	24.8757	25.285	0.9838
grid	24.6649	27.4793	0.8976
hazelnut	37.6479	38.9374	0.9669
leather	26.4941	24.4616	1.0831
metal_nut	34.3597	36.5529	0.94
pill	36.554	37.7873	0.9674
screw	35.9715	37.1229	0.969
tile	25.9535	26.6325	0.9745
toothbrush	36.3193	37.0921	0.9792
transistor	30.3868	32.6993	0.9293
wood	23.5885	28.6674	0.8228
zipper	31.261	32.1882	0.9712

Most categories have an intra-cluster distance value close to 1, suggesting that defective patches are similarly dispersed to normal patches. However, categories such as leather (1.08) show defective patches becoming more diffuse, while wood (0.82) exhibits more compact defective clusters. This suggests that anomalies affect categories differently, with some defects introducing greater variation and others producing consistent visual patterns.

Patch Embedding Projections



The screw category performs noticeably better at the patch level than at the CLS embedding level. With stats being: (This is incorrect to say they are not overly comparable can mention this just word it better)

- Silhouette ≈ 0.125
- Separation ≈ 1.2
- Intra-ratio ≈ 0.97

One possible explanation is that the anomalous regions within screw images typically occupy only a small portion of the image. As a result, the global CLS embedding is dominated by information from the surrounding normal screw structure, causing normal and defective images to appear highly

similar in the image-level embedding space. In contrast, patch embeddings focus on local image regions, allowing patches containing anomalies to be represented separately from normal patches. This suggests that local representations are better able to capture the subtle visual differences associated with screw defects.

Statistical Modelling Algorithms

This section investigates how the algorithm proposed by Tango Dias and Paula Amaral in *A Classification Method Based on a Cloud of Spheres* [2] can be adapted for unsupervised anomaly detection. The original algorithm is first reinterpreted for the unsupervised setting before being extended to use ellipsoidal regions in place of hyperspheres.

Several assumptions made in the original work are not directly applicable to high-dimensional (768-dimensional) Vision Transformer embeddings or to the unsupervised anomaly detection setting. Consequently, a number of modifications to the original algorithm are introduced. These adaptations are motivated throughout this chapter and their influence on both detection performance and interpretability is evaluated experimentally.

Unsupervised Hypersphere Fitting Algorithm

This notebook implements and adapts ideas from *A classification method based on a cloud of spheres* [2]. The original work proposes constructing spherical regions to define decision boundaries between classes. In this project, the method is adapted for unsupervised anomaly detection using MVTec AD and DINOv2 embeddings.

The original paper differs from this setting in three important ways:

- Supervised boundary construction:

The original algorithm constructs spheres using both positive and negative training samples. In the unsupervised anomaly detection setting, only normal training embeddings are available, meaning sphere boundaries must be estimated without any knowledge of defective samples. To address this limitation, candidate regions are initialised from the densest local K-nearest neighbour (KNN) neighbourhood and subsequently expanded using a constrained growth procedure.

- Connected manifold assumption:

The original method assumes that the data lie on a connected manifold and therefore encourages neighbouring spheres to form connected regions. This assumption is less appropriate for DINOv2 embeddings, where normal samples from a single object category may occupy multiple disconnected regions within the 768-dimensional embedding space. As defects are not observed, there is no prior knowledge of where anomalous regions exist. Consequently, the adapted algorithm does not enforce connectivity between neighbouring hyperspheres, allowing disconnected regions of normality to be represented independently.

- Exclusive embedding assignment:

The adapted algorithm enforces that each normal training embedding belongs to exactly one hypersphere. During the growth stage, candidate regions are iteratively cleaned to ensure that previously assigned embeddings are not incorporated into newly generated hyperspheres. While geometric overlap between hyperspheres is permitted, overlap in embedding ownership is not. This results in a unique ownership of normal training embeddings, while allowing neighbouring hyperspheres to share unoccupied regions of the embedding space.

How does Mahalanobis fit to the data

This section visualises how a global Mahalanobis ellipsoid fits the normal training embeddings and uses it as a baseline for comparison with the proposed hypersphere fitting algorithm.

The objective is not to evaluate the performance of Mahalanobis, but rather to understand the assumptions it makes about the embedding space. In particular, Mahalanobis models the normal distribution as a single ellipsoidal region defined by the global centroid and covariance matrix. The following visualisations illustrate how well this assumption represents the underlying embedding distribution before introducing the proposed multi-region fitting approach.

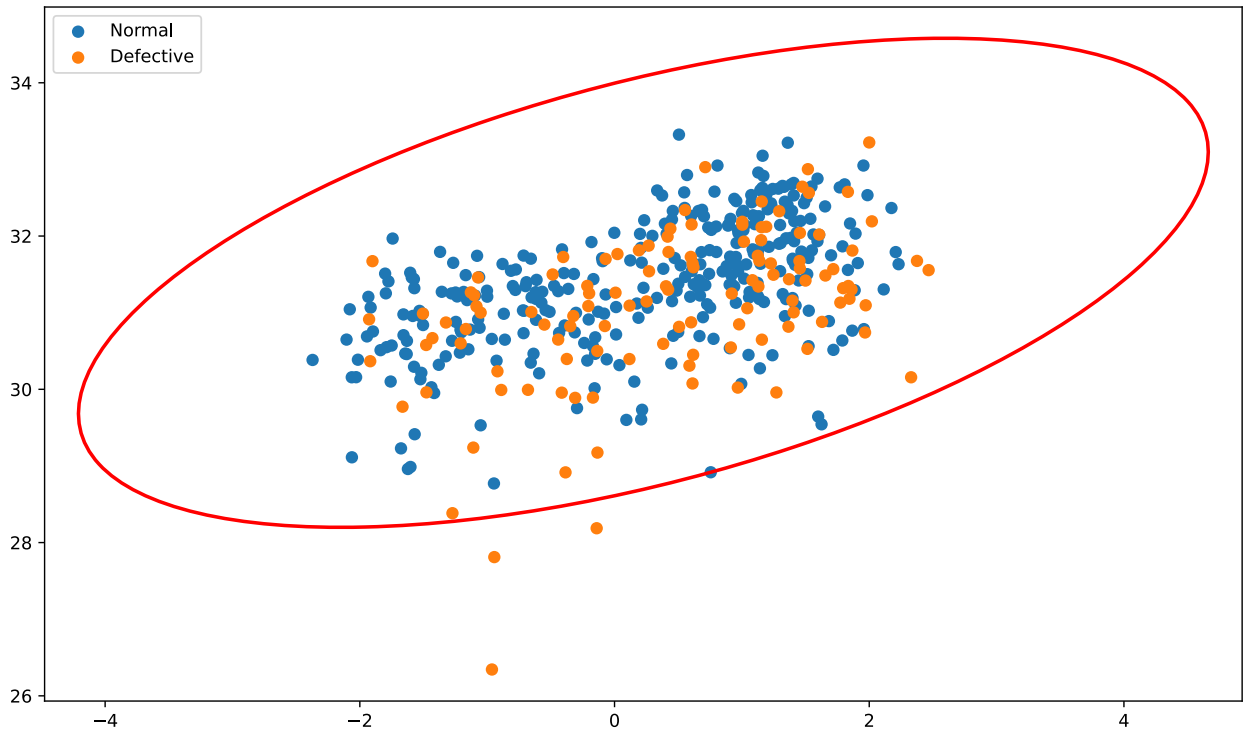


Figure 10: Mahalanobis fit of the screw category in PCA

For the screw category, it can be seen that the Mahalanobis ellipsoid captures the dominant variance of the normal training embeddings along the first two principal components. The orientation of the ellipsoid reflects the covariance structure of the data, demonstrating how Mahalanobis adapts to anisotropic variance rather than assuming a spherical distribution. However, the figure also shows considerable overlap between the projected normal and defective embeddings, illustrating why this category remains challenging for a single global statistical model and contributing to the observed AUROC of 0.802.

Choosing a Value for K

The first step of the proposed algorithm is identifying the densest local region of the normal embedding space. This is achieved using a K-nearest neighbours (KNN) search, where the embedding with the smallest average distance to its neighbours is selected as the initial sphere centre.

Selecting an appropriate value for K is important. If K is too small, the initial neighbourhood becomes overly sensitive to local noise and may fail to capture a representative region. Conversely, if K is too large, the neighbourhood becomes overly broad, increasing the likelihood of creating a hypersphere that encompasses a significant proportion of the embedding space.

As each MVTec AD category contains a different number of normal training samples, selecting a fixed value of K would not scale consistently across categories. Instead, K is defined as a percentage of the available training embeddings. Three candidate values (2.5%, 5%, and 10%) are visually compared to determine a suitable compromise between capturing a dense local neighbourhood while avoiding an overly large initial region.

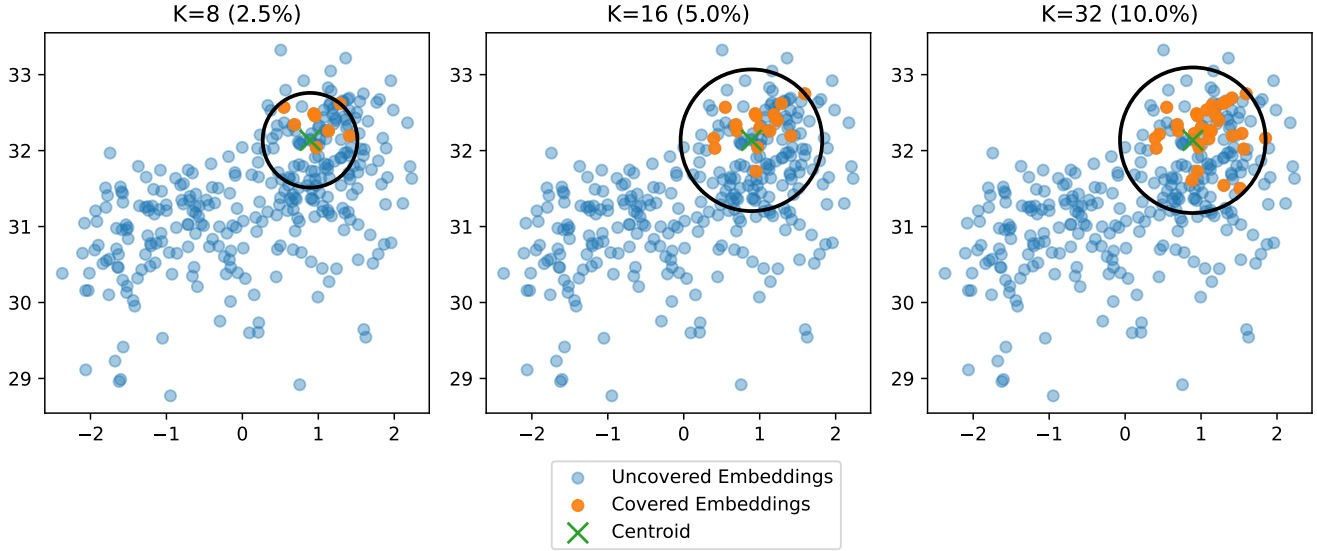


Figure 11: K values for each percentage of the screw category

The value in the end selected was 5% of the available training embeddings. Visually, this provided the best balance between capturing a representative local neighbourhood while avoiding an initial hypersphere that encompassed an excessively large region of the embedding space.

Although this percentage is unlikely to be optimal for every category, as the structure of the embedding space varies between object classes, it provides a consistent and scalable initialisation strategy across the MVTec AD dataset. Furthermore, the subsequent growth stage allows the hypersphere to adapt to the local embedding distribution, reducing the sensitivity of the overall algorithm to the initial choice of K .

Growth Rate for the Hyperspheres

Although the initial KNN neighbourhood provides a suitable starting point, the choice of K remains arbitrary and should not solely determine the size of a hypersphere. To reduce the dependence on this initial neighbourhood, a growth stage is introduced.

Starting from the initial KNN region, the hypersphere is iteratively expanded by a small growth factor. After each expansion, any newly enclosed embeddings are incorporated into the region and the hypersphere is recalculated using the updated centroid and radius. This process continues until no additional embeddings can be added.

The purpose of the growth stage is to allow each hypersphere to adapt naturally to the local structure of the embedding space rather than being constrained by the initial KNN selection. In particular, it reduces the likelihood of producing a large number of singleton or very small hyperspheres, allowing neighbouring embeddings that belong to the same local region to be grouped together.

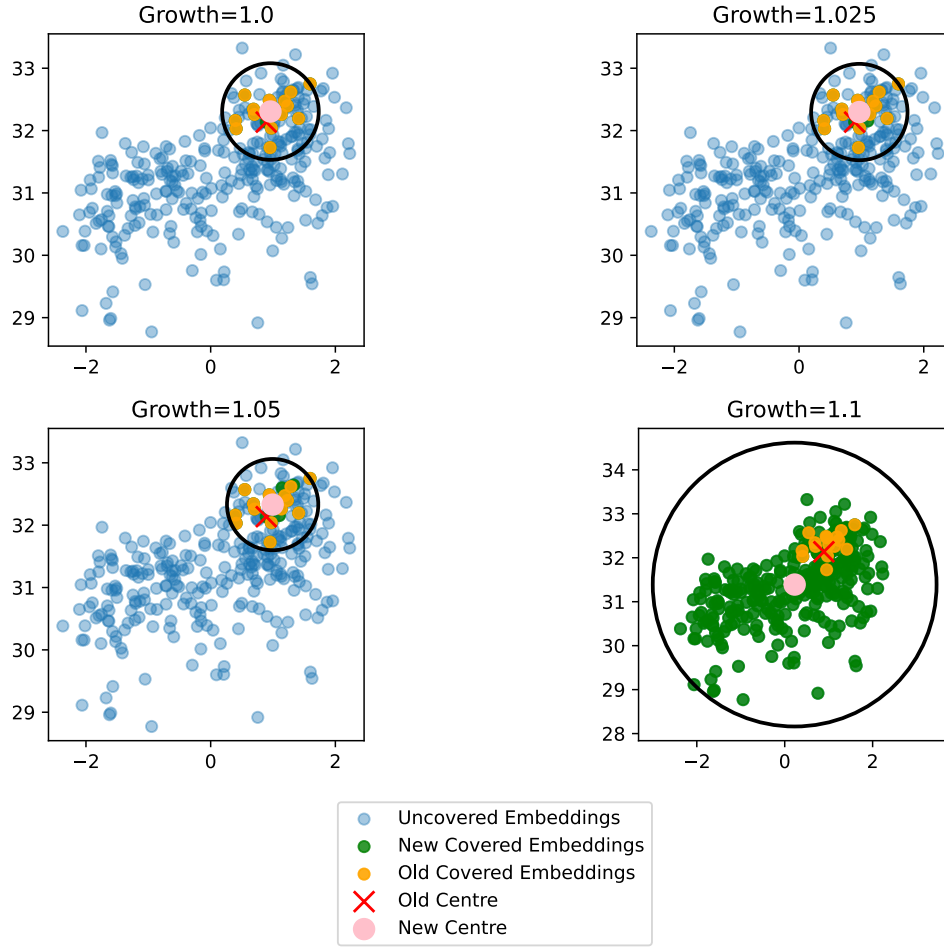


Figure 12: Various growth rates for the screw category

A growth rate of 5% was selected as the most suitable value. This provided enough expansion to include additional nearby embeddings from the same local neighbourhood, while avoiding excessive growth. A smaller growth rate of 2.5% produced little change from the initial KNN region, whereas a larger growth rate of 10% caused the hypersphere to expand too broadly, creating a catch-all region. This behaviour would reduce the benefit of the proposed method, as the resulting hypersphere would begin to resemble a global centroid-based model rather than a local region-based representation.

Adaptive Multi-Hypersphere Fitting

This cell implements the full unsupervised hypersphere fitting algorithm. The aim is to cover all normal training embeddings using a set of local hyperspheres, while ensuring that each training embedding is assigned to only one region.

The algorithm proceeds as follows:

1. **Initialise uncovered embeddings** All normal embeddings begin as uncovered. A boolean mask is used to track which embeddings have not yet been assigned to a hypersphere.
2. **Find the densest uncovered region** For the current set of uncovered embeddings, a FAISS KNN search is performed. The embedding with the smallest average distance to its neighbours is selected as the centre of the densest local region.

3. **Create an initial candidate hypersphere** The densest embedding and its nearest neighbours form the first candidate region. This candidate is cleaned to ensure it does not contain embeddings already assigned to previous hyperspheres.
4. **Grow the candidate hypersphere** If the initial candidate is valid, the hypersphere is expanded using a small growth factor. Newly included uncovered embeddings are added to the candidate region.
5. **Clean after growth** After each growth step, the candidate is cleaned again. If the cleaned candidate contains more embeddings than before, the growth is accepted. Otherwise, growth stops.
6. **Store the final hypersphere** Once no further valid growth is possible, the final centroid, radius, and assigned embedding indices are stored. These embeddings are then marked as covered.
7. **Repeat until full coverage** The process repeats until every normal training embedding has been assigned to a hypersphere.

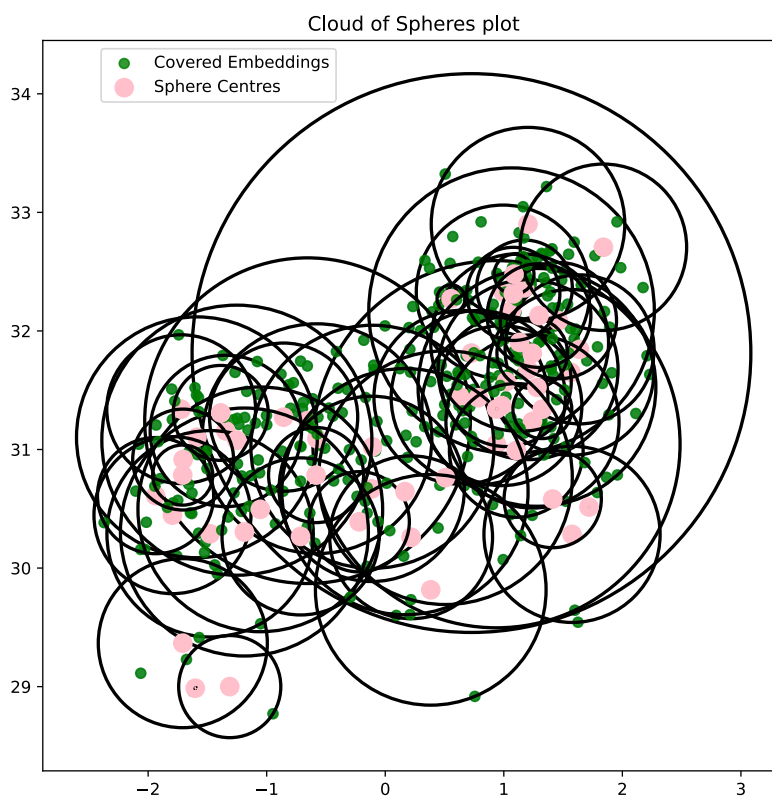


Figure 13: Hypersphere PCA plot for screw category

Unsupervised Ellipsoid Fitting Algorithm

This notebook extends the hypersphere-based algorithm by replacing each spherical region with an ellipsoid. The motivation is that normal embeddings in DINOv2 feature space are unlikely to form locally isotropic clusters. Instead, neighbourhoods may stretch more strongly along some directions than others. Ellipsoids are therefore able to model local variance more naturally than hyperspheres.

The ellipsoid formulation has several advantages:

- Aligns each region with the natural variance structure of the local KNN neighbourhood.
- Reduces unused empty space compared with hyperspheres, since the boundary can contract along low-variance directions.

- It can reduce unnecessary overlap between neighbouring regions by following the dominant principal axes of the local embedding distribution.
- It is better suited to high-variance categories, where the normal embedding space may contain elongated or anisotropic regions.
- Provides additional interpretability through eigenvalues, eigenvectors, axis ratios, and local region structure.

Several changes were introduced compared with the hypersphere version:

- Growth is variance-scaled rather than uniform. Expansion along each axis is controlled by the relative eigenvalue contribution, so high-variance directions can grow more than low-variance directions.
- Candidate cleaning is weight-based. Instead of immediately removing a point when a candidate ellipsoid overlaps a previous region, the algorithm first reduces that point's contribution to the ellipsoid fit. If its weight reaches zero and overlap remains, the point is removed.
- Sparse ellipsoids require additional support. Unlike hyperspheres, ellipsoids fitted from very few points can become geometrically unstable. To address this, the covariance of a small candidate region is blended with covariance information from a previous ellipsoid.

Varaince-Scaled Growth

In the hypersphere algorithm, uniform growth was appropriate because the radius is identical in every dimension. Expanding the hypersphere therefore preserved its geometry regardless of direction. This assumption no longer holds for ellipsoids, where each principal axis represents a different amount of local variance. Local neighbourhoods in the DINOv2 embedding space are often anisotropic, meaning that the variance differs substantially between principal axes. Uniformly scaling every axis would therefore assume that each principal component contributes equally to the local structure of the embedding space. In practice, this can cause low-variance directions to expand excessively, increasing the amount of empty embedding space enclosed by the ellipsoid and the likelihood of overlap with neighbouring regions.

To address this, the proposed algorithm scales growth according to the variance explained by each principal axis. Consequently, axes with greater variance receive proportionally more growth, while lower-variance axes expand more conservatively.

Let the eigenvalues of the covariance matrix be λ_i , where $\lambda_{\max}$ is the largest eigenvalue. The relative variance contribution of axis i is defined as

$$r_i = \max\left(\frac{\lambda_i}{\lambda_{\max}}, r_{\min}\right)$$

A minimum variance ratio $r_{\min}$ is introduced to ensure that every principal axis receives a small amount of growth, preventing axes associated with very small eigenvalues from becoming numerically unstable.

The growth factor for each axis is then:

$$a_i = 1 + gr_i$$

where g is the global growth parameter

The grown eigenvalue is:

$$\lambda_{i'} = \lambda_i a_i^2$$

A point x is inside the grown ellipsoid if:

$$\sum_i \frac{(v_i^T(x - \mu))^2}{\lambda_{i'}} \leq \tau$$

where:

- μ = ellipsoid centre
- v_i = eigenvector for axis i $r_{\min}$ = minimum variance ratio
- λ_i = eigenvalue for axis i $\lambda_{i'}$ = grown eigenvalue
- τ = ellipsoid threshold
- g = growth factor

The result of this allowed the algorithm to grow in direction of where points already are and potentially capturing anymore points into the ellipsoid around its principal axes that KNN was unable to capture.

Candidate cleaning

In the hypersphere algorithm, candidate cleaning was performed by iteratively removing the point that determined the current sphere radius. Since the radius is defined by the furthest point from the centroid, removing this point always reduces the size of the sphere and therefore the likelihood of overlap with previously assigned regions.

This strategy does not naturally extend to ellipsoids. The size and orientation of an ellipsoid are jointly determined by its covariance matrix rather than by a single boundary point. Consequently, removing the embedding with the largest Mahalanobis distance does not necessarily reduce overlap with previously assigned ellipsoids, as that embedding may contribute little to the principal direction responsible for the overlap. Repeated deletion can therefore remove many embeddings before a satisfactory solution is obtained.

To address this, candidate cleaning is rebuilt as a weighted covariance estimation problem. Rather than immediately removing an embedding, the algorithm first identifies the principal axis contributing most to the overlap and progressively reduces the contribution of the corresponding embedding to the covariance matrix. The embedding weight is updated using a binary search until either a valid ellipsoid is obtained or the weight reaches zero, at which point the embedding is removed from the candidate and the covariance matrix is recomputed.

This approach attempts to preserve as much local neighbourhood information as possible before permanently discarding an embedding. However, the current implementation has an important limitation. An embedding whose weight has been reduced to zero may still lie geometrically inside the resulting ellipsoid despite no longer contributing to its covariance estimate. Although this embedding is no longer considered part of the fitted neighbourhood, it may still be enclosed by the final ellipsoid, suggesting that future work should jointly optimise covariance estimation and geometric membership.

Let the candidate neighbourhood be

$$\mathcal{X} = \{(x_i, w_i)\}_{i=1}^n$$

where $w_i \in [0, 1]$ is the weight associated with embedding x_i .

The weighted centroid is given by

$$\mu = \sum_i^n w_i x_i$$

The weighted covariance matrix is then estimated as

$$C = \frac{\sum_i w_i (x_i - \mu)(x_i - \mu)^T}{1 - \sum_i w_i^2}.$$

After eigendecomposition,

$$C = V \Lambda V^T,$$

the ellipsoid is constructed from the eigenvectors V and eigenvalues Λ .

If a previously assigned embedding lies within the candidate ellipsoid, an encroachment has occurred. For an encroaching embedding x_s , the contribution of each principal axis is computed as

$$c_{i(x_s)} = \frac{(v_i^T (x_s - \mu))^2}{\lambda_i}.$$

The principal axis responsible for the greatest contribution is

$$i^* = \arg \max_i c_{i(x_s)}.$$

The candidate embedding contributing most strongly along this axis is then identified by

$$j^* = \arg \max_j \frac{(v_{i^*}^T (x_j - \mu))^2}{\lambda_{i^*}}.$$

Rather than immediately removing this embedding, its weight is reduced using a binary search,

$$w_{j^*} \in [0, w_{j^*}].$$

The covariance matrix is recomputed after each update until either no encroachment remains or the weight reaches zero. If the weight becomes zero and overlap still exists, the embedding is permanently removed from the candidate neighbourhood and the ellipsoid is refitted.

Covariance Support Estimation

Unlike hyperspheres, ellipsoids require a reliable covariance estimate to define their shape and orientation. When only a small number of embeddings are available, the covariance estimate becomes unstable, often producing highly elongated or degenerate ellipsoids. This is particularly problematic during the later stages of the covering algorithm, where only a few uncovered embeddings may remain.

To address this, the proposed algorithm introduces a covariance support mechanism. If the number of embeddings used to fit an ellipsoid falls below a predefined threshold, the covariance estimate is blended with that of a neighbouring ellipsoid. This provides a more stable estimate while preserving the local centroid of the candidate region.

The current implementation selects the supporting ellipsoid based solely on Euclidean distance between ellipsoid centres. This assumes that nearby ellipsoids exhibit similar covariance structure, however this is not necessarily true. A more principled approach would identify the neighbouring ellipsoid that is most similar in shape, for example using eigenvalue ratios or principal axis alignment, rather than spatial proximity alone.

A further limitation is that repeated covariance borrowing can produce multiple ellipsoids with very similar covariance structure. In the extreme case, this may result in neighbouring ellipsoids becoming almost identical in shape. While this would be undesirable for applications requiring accurate local density estimation, it is less problematic for anomaly detection. The objective of the

proposed algorithm is to partition the normal embedding space rather than recover its exact local geometry, and normal training embeddings are expected to exhibit broadly consistent covariance structure.

Let the covariance estimated from the current candidate neighbourhood be

$$C_{\text{local}} = \frac{\sum_i w_i (x_i - \mu)(x_i - \mu)^T}{1 - \sum_i w_i^2}.$$

The supporting ellipsoid is selected as the previously fitted ellipsoid whose centre is closest to the current candidate,

$$k^* = \arg \min_k \|\mu - \mu_k\|_2.$$

Its covariance matrix is reconstructed from its eigendecomposition,

$$C_{\text{support}} = V_k \Lambda_k V_k^T.$$

The final covariance estimate is obtained by blending the local and supporting covariance matrices,

$$C = \alpha C_{\text{local}} + (1 - \alpha) C_{\text{support}},$$

where

$$\alpha = \min \left(1, \frac{n}{n_{\text{support}}} \right),$$

n is the number of embeddings in the current candidate neighbourhood and n_{support} is the minimum number of embeddings required to estimate a stable covariance matrix.

Results

This section presents the experimental evaluation of the proposed hypersphere and ellipsoid algorithms on the MVTec AD dataset. The methods are compared against several statistical baselines using image-level AUROC. The results are analysed both in terms of overall performance across all categories and individual category performance to identify the strengths and limitations of the proposed approaches.

	mean	median	std	Min Category	Value	Max Category	Value
Centroid	0.845	0.849	0.135	screw	0.506	leather	0.997
KNN	0.923	0.933	0.06	screw	0.804	leather	1.0
Avg-KNN	0.92	0.926	0.062	screw	0.786	leather	1.0
Mahalanobis	0.958	0.972	0.049	screw	0.802	tile	1.0
Hypersphere	0.89	0.919	0.083	screw	0.778	tile	0.997
Ellipsoid	0.927	0.93	0.057	screw	0.807	leather	1.0

AUROC for each scoring method

The proposed hypersphere algorithm performs similarly to the Centroid, KNN and Avg-KNN scoring methods, which is expected as these statistics are incorporated during the construction of the hyperspheres. However, the final mean AUROC is lower than the KNN-based baselines. Inspection of the per-category results suggests that, for several categories, the fitted hyperspheres become sufficiently large that their behaviour approaches that of the Centroid method.

As the Centroid baseline achieves the lowest overall performance, this reduces the mean AUROC of the hypersphere algorithm. One possible explanation is that the global growth factor is not equally suitable for every category. Categories with more dispersed/compact embedding distributions may require larger/smaller growth, suggesting that category-specific or adaptive growth strategies should be investigated in future work.

The proposed ellipsoid algorithm achieved the second-best median AUROC, indicating that it performed competitively across most categories. However, a small number of outlier categories reduced the overall mean performance, with the transistor category performing substantially worse than the other methods despite similar performance on categories that were generally considered easier.

A possible explanation for the poor performance on transistor is the covariance support estimation. Unlike most categories, transistor contained relatively few large ellipsoids, with the largest consisting of only 11 embeddings. Covariance support was enabled for ellipsoids containing fewer than five embeddings, however only 18 of the 43 fitted ellipsoids contained enough embeddings to estimate their covariance independently. Consequently, over half of the ellipsoids relied on borrowed covariance from neighbouring regions.

Examining the eigenvalue ratios ($\frac{\lambda_{\max}}{\lambda_{\min}}$) of these ellipsoids shows that many possess extremely anisotropic covariance structures, where a single principal direction dominates the local variance. Repeatedly borrowing these highly anisotropic covariance estimates may therefore cause this dominant direction to propagate throughout the embedding space, stretching neighbouring ellipsoids in directions that are not supported by their own local embeddings. This can cause the normal region to expand excessively, resulting in more defective embeddings being classified as normal and ultimately reducing the AUROC.

	Centroid	KNN	Avg-KNN	Mahlanobis	Hypersphere	Ellipsoid
tile	0.992	0.995	0.997	1.0	0.997	0.997
zipper	0.962	0.971	0.973	0.987	0.78	0.885
toothbrush	0.961	0.933	0.931	0.972	0.783	0.871
hazelnut	0.852	0.922	0.919	0.949	0.931	0.968
wood	0.849	0.875	0.87	0.944	0.942	0.958
grid	0.72	0.953	0.944	0.983	0.852	0.93
cable	0.78	0.891	0.893	0.945	0.997	1.0
leather	0.997	1.0	1.0	1.0	0.919	0.928
metal_nut	0.778	0.941	0.926	0.983	0.829	0.894
screw	0.506	0.804	0.786	0.802	0.778	0.807
bottle	0.991	0.998	0.994	1.0	0.997	0.998
pill	0.828	0.882	0.891	0.95	0.925	0.933
capsule	0.746	0.861	0.858	0.941	0.805	0.867
carpet	0.928	0.965	0.968	0.98	0.849	0.896
transistor	0.781	0.853	0.849	0.935	0.961	0.975

Interestingly, at the category level the ellipsoidal algorithm generally performed similarly to, or slightly worse than, the global Mahalanobis approach until the screw category, where it outperformed all other methods. A possible explanation can be found in the earlier embedding analysis, where the screw category exhibited a distinctive UMAP structure consisting of three well-separated regions.

Although UMAP is only a low-dimensional projection of the embedding space, this separation suggests that the normal screw embeddings occupy multiple disconnected local neighbourhoods rather than a single compact distribution. A global Mahalanobis model attempts to capture these regions using a single covariance matrix, whereas the proposed ellipsoidal algorithm models each local neighbourhood independently. By fitting multiple ellipsoids to the local covariance structure, the algorithm is able to represent these disconnected regions more accurately, resulting in a higher AUROC for the screw category.

Overall, although the proposed ellipsoid algorithm did not consistently outperform the global Mahalanobis baseline in terms of mean AUROC, it remained competitive across the majority of MVTec AD categories. More importantly, unlike global statistical methods, the proposed approach constructs an explicit partitioning of the normal embedding space through multiple local statistical regions. This provides a more interpretable representation of the decision boundary, allowing individual ellipsoids to be inspected, analysed, and related directly to the local structure of the embedding space. The competitive performance obtained in conjunction with this increased explainability suggests that the proposed algorithm has room to be improved.

Future Work

The first direction for future work is to address the limitations identified throughout this work. In particular, the covariance support estimation could be improved by selecting supporting ellipsoids based on both spatial proximity and covariance similarity rather than centre distance alone.

Similarly, adaptive growth strategies should be investigated to allow the growth factor to vary according to the embedding geometry of each category instead of using a single global parameter.

Candidate cleaning could also be extended by reducing the weights of multiple embeddings rather than a single embedding at each iteration. In principle, this would allow the algorithm to modify the covariance structure more smoothly by reducing the influence of all embeddings contributing to the offending principal axis. However, due to the sparsity of the current embedding space, candidate ellipsoids often contain very few embeddings, making such an approach difficult to estimate reliably. If future embedding spaces become more compact and contain larger local neighbourhoods, this strategy may become a more effective alternative.

A more significant direction for future work is the joint optimisation of the embedding space and the proposed statistical modelling algorithm. The current implementation operates on frozen DINOv2 embeddings, meaning that the embedding space has not been learned with the proposed ellipsoid fitting algorithm in mind. Most existing anomaly detection methods optimise representations by encouraging all normal samples to form a single compact cluster. However, this objective may not be optimal for local statistical modelling.

Instead, the representation learning objective could be designed to preserve the natural local covariance structure of the embedding space while encouraging normal embeddings to organise into compact, well-separated local neighbourhoods. Such an embedding space would be better suited to the proposed hypersphere and ellipsoid algorithms, producing more stable covariance estimates, reducing unnecessary overlap between neighbouring regions, and ultimately allowing the representation learning process and the statistical modelling algorithm to be optimised jointly rather than independently.

Bibliography

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